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Learning Nonclassical Receptive Field Modulation for Contour Detection

Qiling Tang, Nong Sang and Haihua Liu

***Abstract*—This work develops a biologically inspired neural network for contour detection in natural images by combining the nonclassical receptive field modulation mechanism with a deep learning framework. The input image is first convolved with the local feature detectors to produce the classical receptive field responses, and then a corresponding modulatory kernel is constructed for each feature map to model the nonclassical receptive field modulation behaviors. The modulatory effects can activate a larger cortical area and thus allow cortical neurons to integrate a broader range of visual information to recognize complex cases. Additionally, to characterize spatial structures at various scales, a multiresolution technique is used to represent visual field information from fine to coarse. Different scale responses are combined to estimate the contour probability. Our method achieves state-of-the-art results among all biologically inspired contour detection models. This study provides a method for improving visual modeling of contour detection and inspires new ideas for integrating more brain cognitive mechanisms into deep neural networks.**

***Index Terms*—contour detection, nonclassical receptive field, modulation mechanism, multiresolution**

I. INTRODUCTION

CONTOUR depicts important structure information including shape and boundary in images. Contour detection is fundamental for a variety of visual tasks such as object recognition [1], surface segmentation [2], saliency measurement [3], occlusion reasoning [4], and scene understanding [5]. Traditional edge detection approaches [6]–[8] focus on examining the discontinuity of low-level image features, which cannot distinguish between significant structures and trivial edges arising from texture fields.

Contour perception has also provoked great interest in the physiological field for exploring the underlying neurophysiological mechanism. The primary visual cortex (V1) is widely believed to play a pivotal role in the contour perception processing [9]–[12]. Many physiological findings [13]–[16] reveal that stimuli outside the classical receptive field (CRF) of V1 neuron can modulate neural activities within the receptive field. This surrounding region beyond the CRF is called the nonclassical receptive field (nCRF), which greatly extends the effective space for visual cortex neurons to receive feature information. The nCRF modulation allows V1 neurons to integrate long-range information from relatively large parts of the visual field and can be interpreted as the neural substrate of different perceptual tasks such as contour integration and figure-ground segregation at the early stage of visual processing.

The response reduction originating from the surrounding nCRF, called inhibitory effect, makes V1 neuron weak or nonresponsive to broad-field homogenous textures but respond vigorously to various textural contrasts [17], [18]. Additionally, the nCRF modulation can enhance the response of V1 neurons when the context is aligned with stimuli within the receptive field to form specific spatial configurations [19], [20], called facilitatory effect. The inhibitory and facilitatory effects play important roles in reducing cluttered edges and standing out well-organized structures. Combined stimulation of the CRF and the surrounding nCRF increases the sparseness, reliability, and precision of neuronal responses [21]. The interaction between CRF and nCRF is an important characteristic of visual information processing.

Motivated by the nCRF modulation mechanisms, a large number of neural computational models have been proposed to detect image contours [22]–[32]. While considerable progress has been achieved, there is still a major challenge to be solved. The nCRF modulation in these works is usually based on the manually designed connection modes such as the difference-of-Gaussian (DoG) function and the butterfly-shaped pattern. It is clearly inappropriate to describe what happens with a simple mathematical function, owing to the complexity and diversity of neural interactions. Neurons do not evolve according to the convenience of subjectively defined functions.

The integration of different scale cues is another important aspect of contour detection because there exist a large range of scales in natural image structures. Studies of natural image statistics have developed a deep insight into the behavior of multiscale structures in natural images [33], [34]. Psychophysical evidence indicates that the multiscale information processing mechanism of visual systems appears to be sensible for image perception [35], [36], [37]. Two methods are usually adopted to characterize the multiscale property: I) employing feature detectors of different receptive field sizes to extract different image details, and II) building multiresolution images by a resampling technique to produce various perceptual ranges of visual field. Multiscale method based on

Q. Tang is with the School of Biomedical Engineering, South-central University for Nationalities, Wuhan 430074, China (e-mail: qltang@mail.scuec.edu.cn).

N. Sang is with the National Key Laboratory of Science and Technology on Multispectral Information Processing, School of Artificial Intelligence and Automation, Huazhong University of Science and Technology, Wuhan 430074, PR China.

H. Liu is with the Key Laboratory of Congnitive Science, State Ethnic Affairs Commission, South-central University for Nationalities, Wuhan 430074, PR China.

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feature detectors have been widely investigated in contour detection neural models [25], [27], [30], which can better describe structure features of different scales and provide superior performance over single-scale settings. However, multiresolution implementation based on the functional architecture of V1 is rarely reported.

In this paper, we develop a multiscale network architecture inspired by the nCRF modulation for detecting contours in natural images. A multiresolution image pyramid is constructed by resampling the original image. In each resolution subnetwork, the first hidden layer simulates the responses of V1 neurons to stimuli within the CRF, the subsequent layer models contextual influence from the nCRF, the modulatory effects are implemented by combining the CRF and nCRF responses in the next layer, and the modulated responses are then assembled into a contour map. An example of this process is illustrated in Figure 1. For the resampled images, a deconvolution layer is used to restore this contour map to its original size, and a multiscale fusion layer is finally employed to learn how to combine different scale outputs into a single contour map.

The proposed model can learn a corresponding modulatory kernel for each feature map obtained by the CRF, rather than hand-designed functions, and thus create adaptive nonclassical surrounds for different V1 receptive fields. The combined effects of the CRF and nCRF indicate that signal responses are not only sensitive to a specific feature detector but also influenced by contextual information. To our knowledge, this is the first work to introduce the nCRF modulation mechanism into a deep learning framework for contour detection. Our results significantly outperform those obtained with other biologically inspired contour models and are very competitive with the state-of-the-art results based on deep networks.

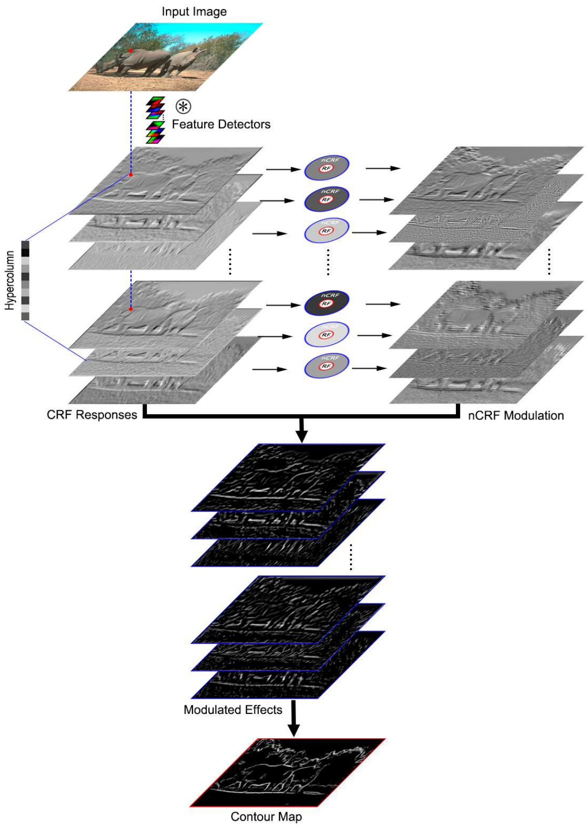

Figure 1. Illustration of the contour detection model inspired by the nCRF modulation. The CRF responses are calculated by convolving the input image with various feature detectors. The hypercolumn at a position is the vector of responses of all units above that position. The nCRF modulatory terms are calculated by convolving the local information with annular modulatory kernels. Different modulated results are combined into a single output.

II. Related Work

The contextual modulation of the nCRF can reduce the responses to homogenous texture fields and enhance the responses to appropriately aligned configurations, which has inspired many efforts to address the problem of eliminating meaningless edges arising from texture fields or cluttered backgrounds and enabling significant structures to stand out from their environments in contour detection [22],[23],[26],[28]. These works have attempted to compensate for the shortcomings of traditional local edge detectors by introducing long-range nCRF influences.

Inhibitory mechanism has been extensively utilized to suppress the responses to texture elements in many neural models [24],[29],[30],[31],[32],[38], in which a half-wave rectification DoG function was treated as an inhibitory surround, and inhibitory strength depends on the homogeneity between the CRF and nCRF areas. In early works, surround homogeneity was measured based mainly on orientation similarity [24], [26]. Contour perception is the coeffect result of properties such as orientation, color, and scale as a result of visual environment complexity [39]. To achieve better inhibitory effects, more models combined various visual cues (such as luminance, color, texton, and sparseness) to calculate homogeneity [29], [31], [32], [40]. In addition, multiscale strategy was introduced to the nCRF inhibition models to further enhance the capability of contour detection under different circumstances [27], [30], [38].

The facilitatory effect is thought to play a role in highlighting perceptually salient structures, and has also been applied to many contour detection models [22],[23],[28],[41],[42],[43], most of which adopt butterfly-shaped modulation surrounds, i.e., inhibitory and facilitatory regions are located at both sides of and along the axis of the receptive field respectively. the facilitatory measure usually follows the Gestalt principle of good continuation. To avoid enhancing redundant responses, the model proposed in [43] enables the facilitatory effect to be much more selective in enhancing the responses to the most likely candidates by combining the sparse representation.

These nCRF-based contour models can significantly reduce texture edges and selectively retain salient structures, improving detection results compared local edge operators. However, there is still much room for improvement in these biologically inspired models.

How can it be determined whether a point is a contour pixel according to multiple cues? A reasonable solution is to exploit the hand-labeled ground truth contours to learn the optimal cue combination. The supervised learning-based methods

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formulate contour detection task as a classification problem. In these discriminative models, the decision for each pixel is usually made by combining multiple features within the image patch centered at the pixel location. For example, the Pb detector uses brightness, color, and texture cues [44]; the BEL detector combines thousands of simple features [45]; and the SCG detector measures contrast by sparse code gradients [46]. The gPb detector further improved patch-based methods by combining multiscale cues defined in the Pb detector into a globalization framework through spectral clustering [47]. This spectral-based globalization strategy was also used by the SCG detector and MCG detector [48]. Recently, the random forest classifier has been successfully applied to contour prediction, such as sketch tokens [49], structured edge (SE) [50] and oriented-edge forests (OEF) [51]. These supervised learning algorithms have led to considerable progress in contour detection. However, these methods are primarily based on hand-designed features, which have limited capability to represent semantically meaningful structures.

In the recent years, deep learning methods have achieved remarkable performances for many applications in computer vision [52],[53]. Deep neural networks have the powerful ability to learn hierarchical feature representations, and these learned features tend to substitute for the manually designed features. Several studies have attempted to apply the deep features, especially learned from convolutional neural networks (CNNs), to the contour detection task and yield state-of-the-art results [54]-[60]. These CNN-based contour detection techniques can be classified into two categories in terms of image training and prediction: patch-to-class, such as N^4-Fields [55], DeepContour [56], and DeepEdge [57]; and image-to-image, including convolutional encoder-decoder network (CEDN) [58], holistically nested edge detection (HED) [59], and richer convolutional features (RCF) [60]. The former takes an image patch as input and predicts the contour probability at the center pixel of the patch by means of a sliding window. A considerable number of overlapping windows densely sampled from the image may result in heavy computational burdens in the training and testing stages. The latter takes a full image instead of individual patches as input and directly yields the contour map of the image as output. This end-to-end scheme originates from fully convolutional neural networks [61]. In addition, deep supervision and side outputs were proposed in [59] and [60], causing more transparent features and coherent contributions for each layer.

Our work presents a novel model of contour detection by embedding the nCRF modulation mechanism into a deep neural network architecture. Unlike previous nCRF-based models, we construct a modulatory kernel for each feature map, thus producing different modulatory behaviors for different feature responses of the receptive fields, and these modulated responses are then integrated optimally, by learning model parameters from labeled data, which avoid burdensome manual design for modulatory surrounds and modulation fusion. To characterize spatial structures at various scales, a multiresolution technique is used to represent visual field information from fine to coarse; a fine scale is beneficial for capturing high-frequency details, while a coarse scale can expand the extent of the receptive fields. The multiscale scheme can further improve contour detection results.

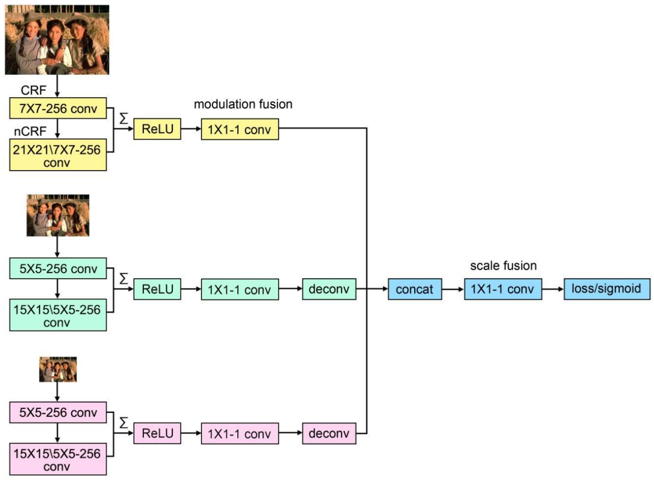

Figure 2. The overall architecture of this pipeline, in which $m\times m\backslash n\times n$-$c$ denotes a homocentric square window with outer size $m\times m$ and inner size $n\times n$ and c is the feature channel number. The network takes as input a multiresolution image pyramid. CRF and nCRF kernels are used to compute the responses within the receptive fields and the modulatory terms from the surrounding regions, respectively. The modulation maps of different sizes are fused to produce a contour possibility map.

III. Nonclassical Receptive Field Modulation Network

In this section, we describe how to construct the network model embedding multiscale nCRF modulation for detecting contours in natural images and discuss the different roles of CRF and nCRF kernels in visual information processing. Then, we introduce the loss function for this task.

A. Network Architecture

Visual processing may be conceptualized as what Helmholtz called the unconscious inference [62], and contextual information and prior knowledge about the object play an important role in this inference process. It is desirable to combine signals from the local receptive fields with contextual information from the surround to carry out the inference of contours in complex scenes. Therefore, we incorporate the nCRF modulation with a deep network framework to build a biologically inspired contour detection model.

The overall architecture of this pipeline is illustrated in Figure 2. Our network takes as input an image of arbitrary size and outputs a contour map of corresponding size. A multiresolution pyramid is constructed to detect image features across multiple scales by subsampling the original image with downsampling factor of 2. The proposed network consists of CRF layers, nCRF layers, modulation fusion layers, deconvolution layers and a scale fusion layer.

The CRF convolutional kernels are utilized to describe the response characteristics of V1 neurons to stimuli within the CRF, and then an annular convolutional kernel outside the CRF is constructed to simulate the nCRF modulation surround for each CRF response map. Neurons respond selectively to only a

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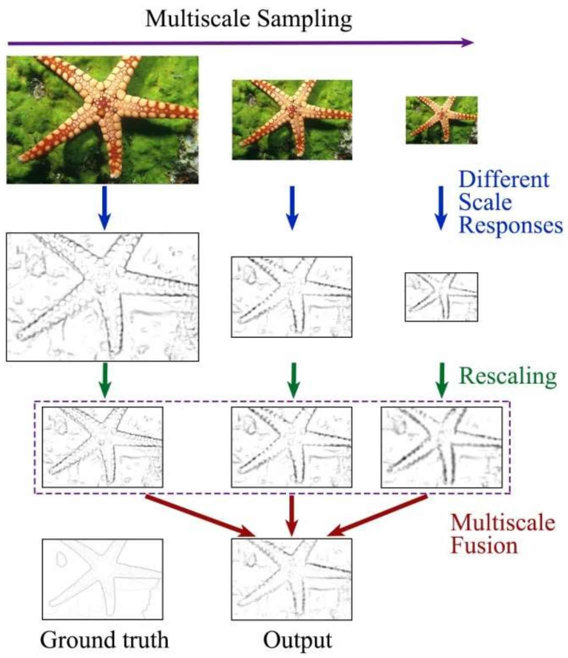

Figure 3. Illustration of our multiscale fusion scheme. A multiresolution image pyramid is built through resampling the input image. Different scale images are fed into the corresponding subnetworks to produce contour responses. Bilinear interpolation is employed to restore these contour maps to the size of the original image, and these maps are then combined into the final result.

small part of input signals at the same time, and a large number of signals are deliberately shielded, which can improve the learning accuracy and extract sparse features better and faster. Accordingly, a rectified linear unit (ReLU) operation is used to perform the half-wave rectification, such that negative values are set to zero. The modulatory effects are achieved by integrating the modulatory properties of the nCRF with established properties of the CRF, followed by a ReLU operation, inducing sparsity on the responses of neurons. Then, an 1×1-1 conv layer fuses various modulated responses at every pixel location into a single result. The modulated response is formulated as follows:

$$R_j^{mod} = ReLU(R_j^{crf} + R_j^{ncrf}) \tag{1}$$

in which R_j^{crf} and R_j^{ncrf} are respectively the CRF response in the j-th feature channel and the corresponding modulatory term coming from its surrounding nCRF, which are computed by

$$\begin{aligned} R_j^{crf} &= \sum_i I_i * K_{i,j}^{crf} + b_j^{crf} \\ R_j^{ncrf} &= R_j^{crf} * K_j^{ncrf} \end{aligned} \tag{2}$$

where $K_{i,j}^{crf}$ and K_j^{ncrf} represent the CRF convolutional kernel of the j-th feature detector over the i-th color channel of input image I and the corresponding nCRF modulatory kernel, respectively. b_j^{crf} is the bias term. We use 256 feature channels in the CRF and nCRF layers in this work.

In order to model V1 responses to various scale features, we employ a multiresolution image pyramid to investigate the modulatory effects at different scales. Figure 3 illustrates our multiscale scheme for contour detection, similar to those proposed in [48], [60]. For the downsampled images, deconvolution layers are inserted to resize the modulation fusion results to the size of the original image using bilinear interpolation. Although we can also learn the parameters of deconvolution layers, there is no noticeable improvements in our experiments. The modulation maps of different sizes are then merged to form a unified result in the scale fusion layer. Finally, a cross-entropy loss / sigmoid layer is used to compute the value of the objective function / the output of the contour probability.

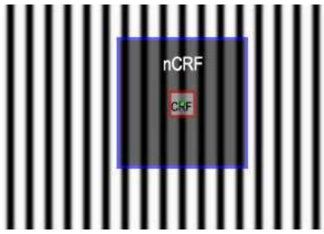

Figure 4. The nCRF modulatory surrounds beyond the CRF allow V1 neurons to integrate broader information from relatively large parts of the visual field.

B. CRF and nCRF Details

The CRF kernels are similar to the convolutional kernels in CNNs, which can selectively respond to different characteristics in terms of local information within the receptive fields. Good initialization of the parameters is helpful for network training.

Gabor function [63], [64] and sparse coding [65], [66] are usually used to mimic the simple-cell receptive field properties of V1 neurons in visual computing models. Recently, sparse autoencoder (SAE) technique [67], [68] is popular for unsupervised feature learning, which can fully utilize large amounts of unlabeled data to capture good feature representations and does not require much prior knowledge about the data, so that the method can be generalized to different cases without making significant modifications. SAE can learn feature representation similar to sparse coding atoms and Gabor filters, and its different hidden units can detect edges at different positions and orientations in images. Therefore, it is reasonable to use a pretrained SAE to initialize the parameters of the CRF layer in our network.

For each resolution images, we randomly sample 500,000 local image patches of the same size as the CRF convolutional kernels from the Berkeley Segmentation Dataset [47] to serve as SAE training data, and the number of SAE hidden units remains the same as that of the CRF layer. The CRF convolutional kernels initialized by the pretrained SAE network are sensitive to edges at different orientations and different positions. The outputs of all CRF units above a given input position are grouped together into one vector (as shown in Figure 1) that is similar to the physiological term "hypercolumn", consisting of V1 neurons responding to the same spatial location in the retina but with different orientation preferences [9], [69]. The CRF kernels utilize local

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Figure 5. Examples of contour detection illustrating the effects of different scales on the contour results. In each row from left to right we present: origin image, ground truth contour, contour detection results obtained by three scales from fine to coarse. For a given image, the receptive fields can capture more local information from its fine resolution setting, and more global information from its coarse resolution setting.

neighborhoods to predict the response to the receptive field center, and the response strength reflects the degree of correlation between the kernel and local information.

The role of the nCRF layer is to produce a corresponding modulatory effect for the response obtained by each CRF kernel. The surrounding region outside the CRF is defined as the nCRF modulation surround. The neurophysiological evidence showed that the extent of the nCRF is normally 2 to 5 times size of the CRF [14], and the size ratio of nCRF to CRF is set to 3 in this work. For example, for a CRF of size 5×5, the corresponding nCRF area is a homocentric square window with an outer size 15×15 and an inner size 5×5. Subsequently, the CRF original responses and the nCRF modulatory terms are accumulated. The joint effects of the CRF and nCRF activate a larger cortical area and thus allow V1 neurons to integrate a broader range of visual information.

Our nCRF kernels can capture long-range contextual information beyond the CRF, which is fundamentally different from the second convolution operation presented in [59], [60] that still acts within the CRF. Although the nCRF alone is unresponsive to visual stimuli, it can integrate information from relatively large parts of the visual field to exert modulatory

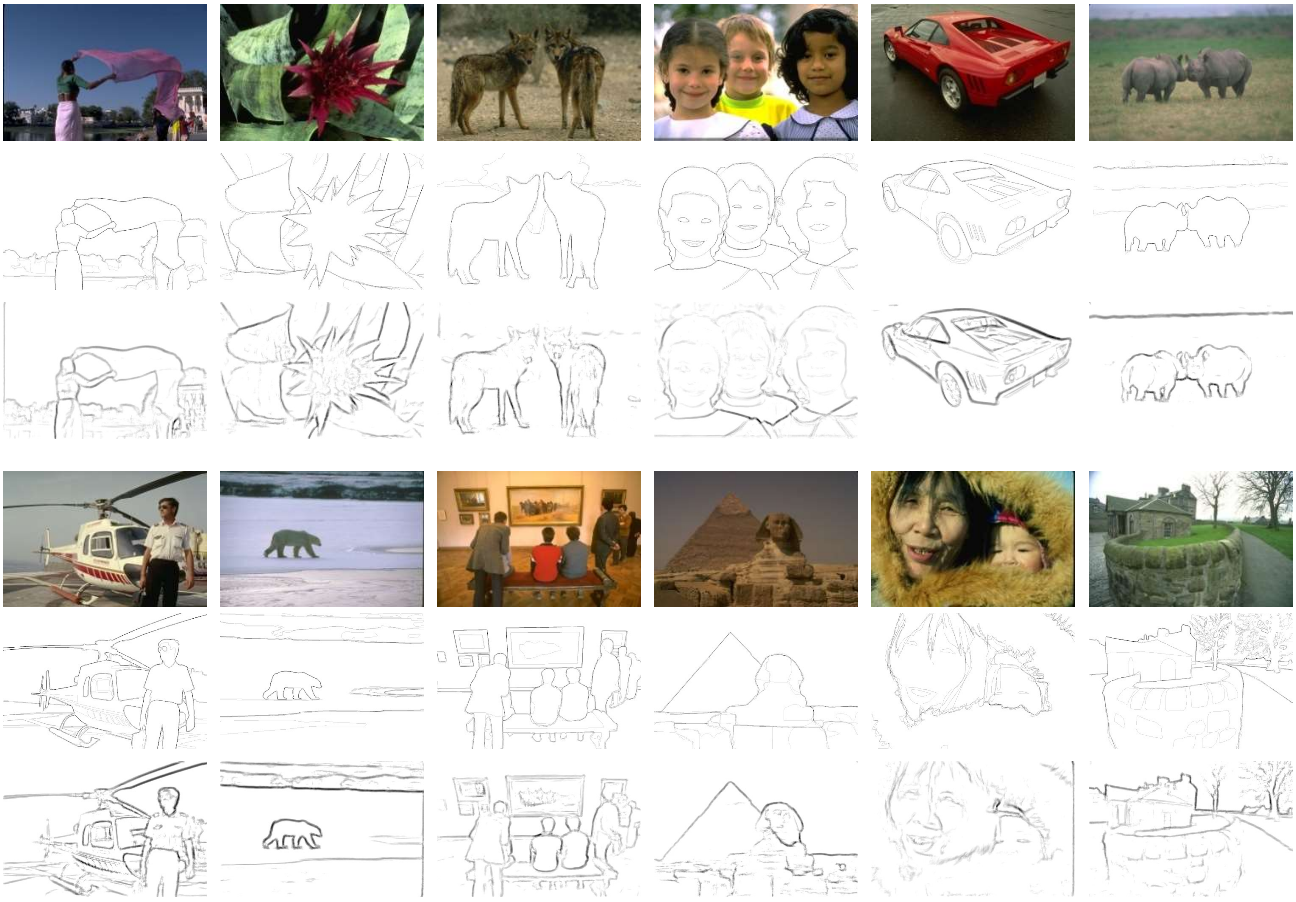

Figure 6. Example results produced by multiscale fusion. Panels from top to down correspond to origin images, ground truth contours, and contour detection results obtained with multiscale modulatory scheme.

effects on the response to the stimuli in the CRF. A synthetic image is used to illustrate the influence of the nCRF modulation on the CRF response, as shown in Figure 4. At the CRF center, a strong CRF response is presented owing to local brightness change in the receptive field (marked with a red box), while the nCRF modulation would weaken the response due to surround inhibition resulting from homogenous texture fields (marked with a blue box). Compared with the effect of CRF alone, the combined effects of CRF and nCRF can integrate information over larger regions of visual space to recognize more complex cases.

C. Loss Function

The proposed network takes advantage of the nCRF modulation mechanism and multiresolution sampling strategy to perform image training and prediction by means of an image-to-image method proposed in [59],[60]; that is, the network takes as input an image and directly outputs contour map of this image. Our goal is to achieve contour results approaching the ground truth through learning the nCRF modulatory effects. In the discriminative model, the contour detection task is transformed to a binary classification problem, i.e., predicting whether a pixel belongs to contour or noncontour. We borrow the cross-entropy loss function defined in [59], which introduces the tradeoff parameters to correct the imbalance in the class distribution of contour and noncontour pixels and is formulated as follows:

$$J = -\sum_i [\beta \cdot \mathbf{1}(y_i = 1)\log \mathrm{P}(y_i = 1 \mid X_i, W) + (1-\beta)\cdot \mathbf{1}(y_i = 0)\log \mathrm{P}(y_i = 0 \mid X_i, W)] \tag{3}$$

where W denotes the collection of all the parameters in this network. Each pixel i of image X has a corresponding ground truth label $y_i \in \{0,1\}$. $\mathbf{1}(\cdot)$ is the indicator function, and $\mathrm{P}(y_i)$ is the probability of X_i belonging to the ground truth label, obtained by the standard sigmoid function. $\beta = |Y^-| \big/ \left(|Y^+| + |Y^-|\right)$ is a controlling parameter to offset heavy class imbalance between contour and noncontour pixels, $|Y^+|$ and $|Y^-|$ denote positive and negative sample sets, respectively. Minimizing the objective function can achieve the maximal probability of the correct class. We use stochastic gradient descent with backpropagation to train this system. In the training procedure, each minibatch contains 10 images, the weight decay is 0.001, and the learning rate and momentum are set to 1e-5 and 0.95 respectively.

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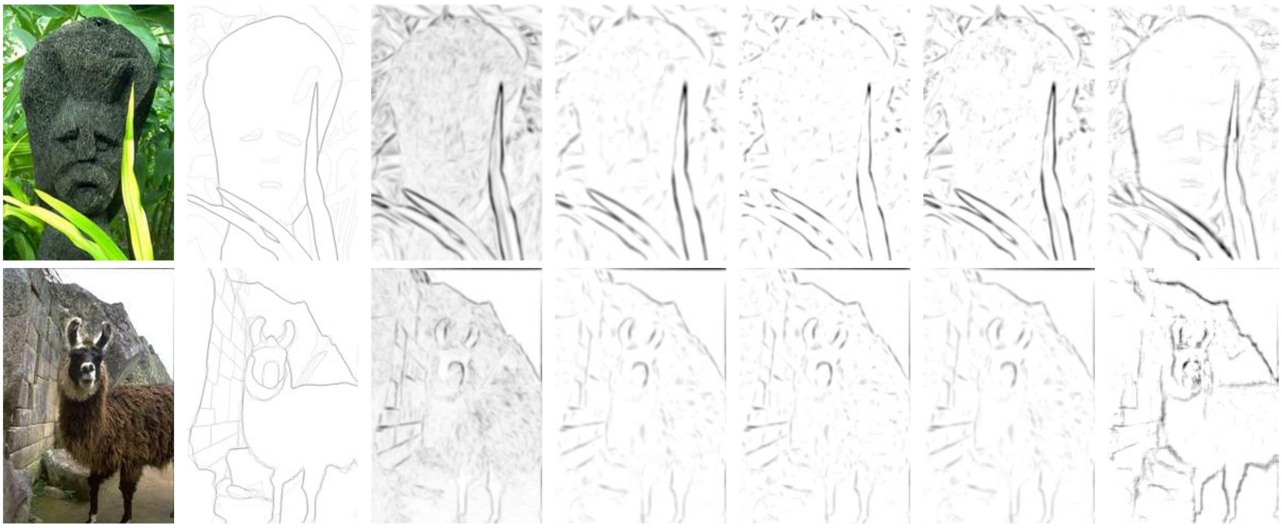

Figure 7. Visual comparison with other contour detection models based on nCRF modulation. From left to right: input image, ground truth, Gabor energy map, Multiscale integration [38], Multifeature-based [31], Contrast-dependent [29], and our results.

IV. Experiments

In this section we report the performance of the proposed network, and compare contour detection results to other biologically inspired models and the state-of-the-art approaches. Contour detection performance is quantitatively evaluated using three standard quantities: the best F-measure on the entire dataset for a universal fixed scale (ODS), the aggregate F-measure on the dataset for the best scale in each image (OIS), and the average precision (AP) on the full recall range [47].

A. Results on the BSDS500 Dataset

We test our network on the popular Berkeley Segmentation Dataset and Benchmark (BSDS500) [47], which consists of 500 natural images of resolution 480×320 (including 200 training, 100 validation, and 200 test images) and the associated ground truth contours manually labeled by several human subjects. The training and validation sets are used for training the network parameters, and the test set is used for the performance evaluation. Data augmentation has proven to be a useful technique for deep network training, and the rotation and flip strategies in HED [59] are utilized to augment the dataset. We initialize the CRF layer with a pretrained SAE net and the nCRF and fusion layers with random values. We find that learned CRF parameters provide very little improvements in our experiments. A feasible interpretation is that the CRF layer learns local edge feature and this low-level feature representation does not depend on specific tasks; thus, unsupervised learning can achieve the desired effect. Consequently, we can fix the CRF parameters and optimize only other parameters during network training.

To verify the benefits of the multiscale nCRF modulation strategy on contour detection, we first investigate the influences of different resolutions on the contour results. Several example results on heavy texture images that are deliberately chosen from the BSDS500 dataset are shown in Fig.5, from which it can be observed that the fine-resolution scheme preserves more detailed object contours and texture edges, while the coarse-resolution scheme is more robust to texture and clutter and tends to detect global significant structures. Coarse-resolution images produced by downsampling can provide a larger extent of visual cues for the same size receptive fields than fine-resolution images; thus, prediction on coarse-resolution images takes more global image contents but may blur the image details, resulting in some missing structures, especially for weak responses.

Combining predictions from multiple resolutions can achieve better performance. Multiresolution result fusion is included as part of our network, and thus, the outputs from different resolutions can be optimally combined by a supervised method, which differs from the multiscale fusion strategy of averaging the contour maps at various resolutions to obtain a final result adopted in RCF [60], where multiscale fusion is not contained in the network architecture but is performed separately. Some example results are shown in Figure 6, from which it is observed that the multiscale scheme makes an excellent tradeoff between reducing texture edges and retaining object structures and presents better visual effects than single-scale schemes.

Figure 7 shows two example results obtained by various contour detection models based on nCRF modulation, including contrast-dependent [29], multifeature-based [31], and multiscale integration [38]. The previous models first utilize the Gabor energy operator to compute the responses of the receptive fields and then modulate the responses according to contextual information. It is observed from Figure 7 that Gabor energy cannot provide good local features and correct orientation cues in cluttered backgrounds for further modulation behavior, which results in these models failing to distinguish between perceptually important structures and trivial local edges, thus reducing texture backgrounds while also destroying image contours. However, our proposed model still exhibits excellent ability to suppress cluttered texture edges and highlight perceptually salient contours.

A standard non-maximal suppression procedure [7] is performed to thin edge maps for evaluation. Table I shows the quantitative evaluation results on each single scale and multiscale fusion. For a single scale, the medium-scale scheme performs best and the coarse-scale scheme achieves relatively low performance in our experiments. Original images can provide more details, which is beneficial to the completeness of the detected contours but produces more trivial edges. Subsampling images blur details, resulting in good performance for coarse-scale structures but poor detection of fine-scale objects. It is desirable to adaptively select an appropriate scale for each pixel location in terms of image content; however, determining this scale is difficult using only bottom-up cues. A widely adopted strategy is to make a prediction at every pixel by integrating the multiple cues across the scale range using supervised learning. The joint effects from different scales can enhance the descriptiveness and discriminative ability of feature representation and yield better classification performance both in F-score and AP compared to the single-scale setting.

In addition, to embody the performance gain of nCRF modulation on contour detection, we also evaluated the results obtained without the nCRF layer. In the absence of a modulation mechanism, the network is only a multiscale local edge detector responding to information within only the receptive field, and the result is unsatisfactory, which suggests that the nCRF modulation mechanism plays an important role in contour processing.

We compare our contour detection method against competing methods: biologically inspired models, including multiscale integration [38], color-opponency with spatial sparseness constraints (SCO) [40], contrast-dependent [29], multifeature-based [31], PC/BC-V1+lateral+texture [43], and

TABLE I. PERFORMANCE OF SINGLE SCALE AND MULTISCALE COMBINATION ON THE BSDS500 DATASET. NOTE THAT THE NETWORK WITHOUT NCRF LAYER OBTAINS A LESS SATISFACTORY RESULT, INDICATING THE MODULATION MECHANISM IS BENEFICIAL TO ACHIEVING DESIRED CONTOUR MAPS.

Method	ODS	OIS	AP
Fine scale	.726	.745	.748
Medium scale	.737	.753	.764
Coarse scale	.635	.649	.617
Without nCFR	.638	.655	.672
Multiscale fusion	.762	.778	.809

TABLE II. CONTOUR DETECTION RESULTS ON THE BSDS500 DATASET

	Method	ODS	OIS	AP
Biologically inspired models	Our Model	.762	.778	.809
	Multiscale integration [38]	.680	-	-
	SCO [40]	.670	-	-
	Contrast-dependent [29]	.630	-	-
	Multifeature-based [31]	.620	-	-
	PC/BC-V1+lateral+texture [43]	.610	-	-
	Adaptive inhibition [30]	.580	-	-
Deep learning based methods	RCF [60]	.811	.830	-
	HED [59]	.788	.808	.840
	CEDN [58]	.788	.804	-
	DeepContour [56]	.756	.773	.797
	DeepEdge [57]	.753	.772	.807
	N^4-Fields [55]	.753	.769	.784
	DeepNets[54]	.738	.759	.758
Non-deep-learning algorithms	OEF [51]	.749	.772	.817
	SE [50]	.746	.767	.803
	MCG [48]	.744	.777	-
	SCG [46]	.739	.758	.773
	Sketch Tokens [49]	.727	.746	.780
	gPb-owt-ucm [47]	.726	.757	.696

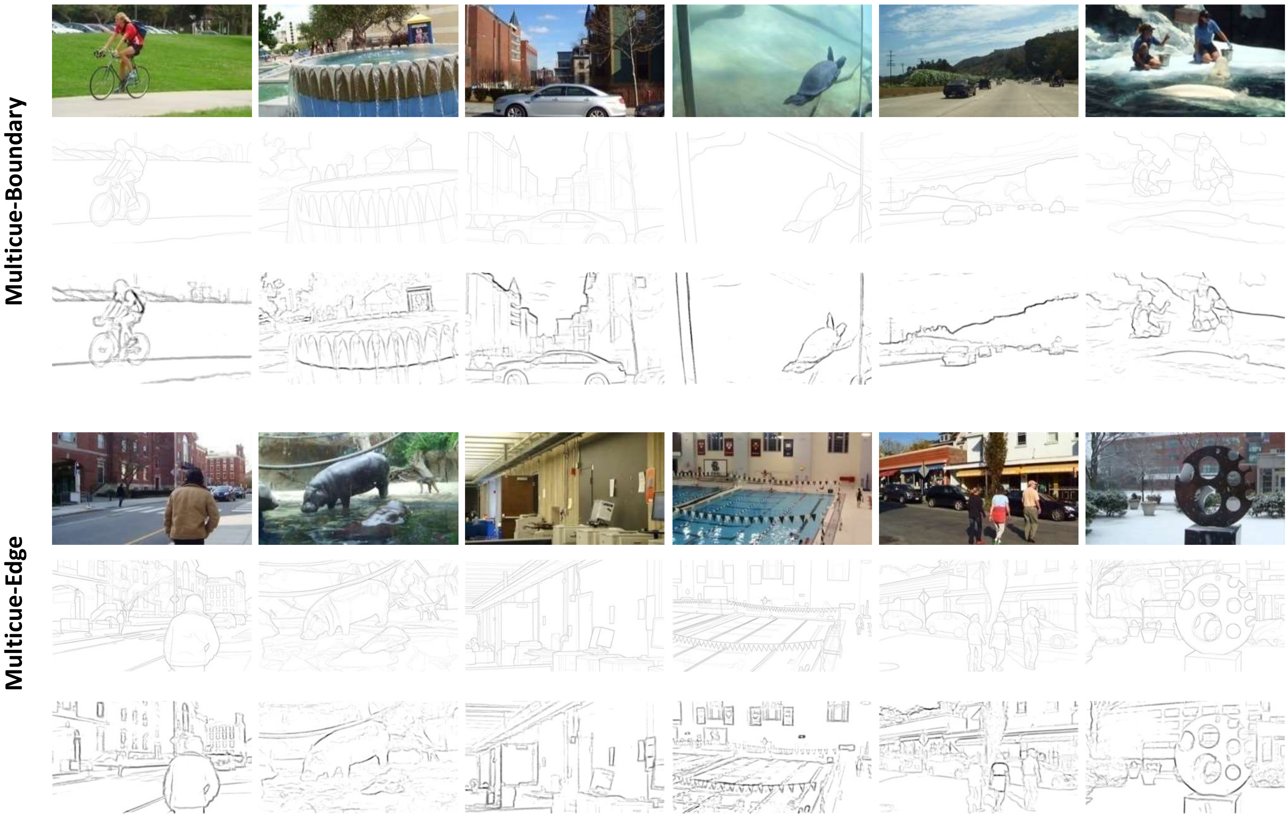

Figure 8. Some example results on the Multicue database. From top to bottom: original images, ground truth contours, and the results produced by our nCRF modulation method.

adaptive inhibition [30]; deep learning-based methods, including RCF [60], HED [59], CEDN [58], DeepContour [56], DeepEdge [57], N^4-Fields [55], and DeepNets [54]; and other learning-based (nondeep-learning) algorithms, including OEF [51], SE [50], MCG [48], SCG [46], sketch tokens [49], gPb-owt-ucm [47].

The quantitative comparison with the competitors on the BSDS500 dataset is summarized in Table II. Our approach achieves a substantial improvement over other biologically inspired contour detection models, which is mainly due to the learned modulatory kernels and multiresolution information processing scheme. This proposed network mainly simulates the activity of neurons in V1 and thus is a shallow architecture. The performance of our method is below those of state-of-the-art methods based on deeper networks, such as RCF [60], HED [59], and CEDN [58], some of which even outperform the result of human visual perception for contours (ODS F-measure 0.803). However, the network architecture of these methods is much deeper than ours; for example, RCF [60] and HED [59] are built on a VGG 16-layer net [70], which indicates that high-level object information produced by deep networks is indeed beneficial for improving contour detection results, especially in complex natural images. Although there is no high-level object information in our network, the results are very encouraging for a shallow network.

B. *Results on the Multicue Dataset*

To investigate how to optimally combine visual cues for boundary detection, Mély et al. [71] recently constructed a multicue edge and boundary dataset that contains 100 short binocular video clips of natural scenes captured by a stereo camera. For each video clip, the last frame of the left sequence is manually labeled for two annotations, object boundaries and low-level edges. The edge and boundary in the annotation sets are defined according to visual perception at different stages, in which edges arise from an abrupt change in low-level image features that are modeled in the early stages of visual processing, and boundaries represent high-level semantics that are modeled in the later stages of visual processing. The scenes in the Multicue dataset present more complexity than the images in the BSDS500 dataset.

Since the image resolution in the Multicue dataset is very high, we randomly crop 500×500 subimages from original images of resolution 1280×720 during training. As done in the works [59], [60], [71], we randomly choose 80 images for training and 20 images for testing from the dataset, and Multicue-boundary and Multicue-edge are trained separately in terms of the two sets of manual annotations. Some qualitative results on Multicue-boundary and Multicue-edge images are presented in Figure 8.

The evaluation results are obtained by averaging the scores over three independent runs. The results of various approaches are reported in Table III, including human subjects, the method proposed by Mély et al. [71], HED [59], and RCF [60]. Our method performs significantly better than human subjects and

TABLE III. BOUNDARY AND EDGE DETECTION RESULTS ON THE MULTICUE DATASET

Method	ODS	OIS	AP
Human-Boundary [71]	.760 (.017)	-	-
Multicue-Boundary [71]	.720 (.014)	-	-
Our Model	.799 (.012)	.807 (.007)	.841 (.013)
HED-Boundary [59]	.814 (.011)	.822 (.008)	.869 (.015)
RCF-Boundary [60]	.825 (.008)	.836 (.007)	-
Human-Edge [71]	.750 (.024)	-	-
Multicue-Edge [71]	.830 (.002)	-	-
Our Model	.843 (.008)	.855 (.009)	.872 (0.12)
HED-Edge [59]	.851 (.014)	.864 (.011)	.890 (.007)
RCF-Edge [60]	.860 (.005)	.864 (.004)	-

that proposed by Mély et al., and slightly worse than HED and RCF. In edge detection task, the ODS F-measure of our method is lower than the highest F-measure reported in RCF-Boundary [60] by a small margin of 1.7%, and in boundary detection task, our performance is lower by a larger gap of 2.6%, which indicates that high-level feature learning plays a more important role in detecting semantic object boundaries. The reason for the difference between the performance of edge detection and that of boundary detection is that our nCRF modulation is an early visual activity that occurs in primary visual cortex and does not involve later visual cortex, so there is a lack of the ability to learn high-level features. Determining how to build a deeper network to capture more high-level features is also the goal of our further work.

V. CONCLUSION

This work presents a novel network architecture inspired by modulation behavior in V1 for contour detection in natural images. Our model achieves significant advantages over other biologically inspired contour detection models, mainly due to two aspects: I) the nCRF modulatory kernels are obtained by learning; thus, the resulting modulation behaviors can adapt well to the response maps of the receptive fields and can effectively integrate information over larger regions of visual space to recognize complex features; II) the multiresolution technique is used to represent visual field information from fine to coarse, and the combined effects from different resolutions enhance the descriptiveness and discriminative ability of feature representation. This study provides a method for improving the bio-inspired computational model of contour detection and inspires new ideas for integrating more brain cognitive mechanisms into deep neural networks.

The proposal mainly focuses on modulation mechanism in V1. Although V1 contributes considerably to contour perception, contour processing in the visual system is rather complex and is not limited to the V1 area. Physiological and psychophysical results indicate that some higher cortical areas (e.g., V2, V4) also participate in this visual task, and it has been widely believed that contour detection in cluttered scenes involves the visual perception of various cortical areas [72],[73]. Furthermore, there is important evidence of the presence of feedback loops from higher areas to lower-level sensory regions in contour integration [74],[75]. How to integrate higher cortical areas and recurrent feedback mechanisms into this contour detection network model will be the goal of our further study.

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